User Guide: Mores Creek Summit UAVSAR/Lidar Datacube

`ZARR/mores_creek_summit.zarr` and the modeling-team subset `mcs_top_pairs_subset.nc`

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September 2026

1 Overview

The datacube collects every SnowEx UAVSAR interferometric product and airborne lidar product over Mores Creek Summit (MCS), Idaho, on one shared analysis grid, together with the derived quantities needed to turn interferometric phase into snow water equivalent change (ΔSWE): local incidence angle, atmospheric phase delay, and the ΔSWE retrievals themselves.

- **Format:** Zarr v2, consolidated metadata (9.4 GB on disk, 15 GB logical).
- **Grid:** 3 m, UTM zone 11N (EPSG:32611); 2801 rows $\times$ 2624 columns; affine transform $(3, 0, 601558 \,/\, 0, -3, 4870872.5)$; derived from the native 0.5 m MCS lidar grid. Northing (`y`) is stored north $\rightarrow$ south.
- **Bounds (lon/lat):** -115.7352 to -115.6355, 43.9072 to 43.9840.
- **Missing data:** `NaN` everywhere, including outside each product's footprint.

2 Dimensions and coordinates

name	size / dims	dtype	meaning
flight (dim)	20	–	unique flight index (line, date)
y (dim)	2801	–	UTM northing of cell centers (m), stored north to south
x (dim)	2624	–	UTM easting of cell centers (m, EPSG:32611)
pair (dim)	22	–	InSAR pair index (order of uavsar_pair_manifest_mcs.json)
lidar_time (dim)	13	–	MCS_Lidar acquisition date (DTM/DSM/CHM)
pol (dim)	4	–	polarization (HH, HV, VH, VV)
line (dim)	2	–	UAVSAR flight line id (05208 NE-looking, 23205 SW-looking)
sd_time (dim)	15	–	snow-depth acquisition date
flight_date	coord on flight	str(10)	acquisition date of each flight
flight_line_id	coord on flight	str(5)	flight line of each flight
heading	coord on pair	int64	aircraft heading of each pair (deg)
lidar_time	coord on lidar_time	datetime	MCS_Lidar acquisition date (DTM/DSM/CHM)
line	coord on line	str(5)	UAVSAR flight line id (05208 NE-looking, 23205 SW-looking)
line_id	coord on pair	str(5)	flight line of each pair
pair_date1	coord on pair	datetime	pair reference (earlier) acquisition date
pair_date2	coord on pair	datetime	pair repeat (later) acquisition date
pol	coord on pol	str(2)	polarization (HH, HV, VH, VV)
sd_source	coord on sd_time	str(14)	snow-depth product (SNEX20_QSI_SD or SNEX_MCS_Lidar)
sd_time	coord on sd_time	datetime	snow-depth acquisition date
temporal_baseline_days	coord on pair	int32	days between the pair's acquisitions
x	coord on x	float64	UTM easting of cell centers (m, EPSG:32611)
y	coord on y	float64	UTM northing of cell centers (m), stored north to south

Two time axes coexist deliberately: `sd_time` (15 snow-depth dates, 2020–2025, QSI + MCS_Lidar) and `lidar_time` (13 MCS_Lidar dates, 2022–2025, the only collection providing DTM/DSM/CHM). The 22 pairs and 20 flights cover winters 2019–20 (line 23205 only) and 2020–21 (both lines).

3 Data variables

variable	dims	dtype	chunks	units / source
atm_delay	(flight, y, x)	float64	(3, 351, 328)	radians; ERA5 delay below aircraft altitude, per flight (compute_atm_delay.py)
atm_delay_diff	(pair, y, x)	float64	(3, 351, 328)	radians; per-pair delay difference (date2−date1) (compute_atm_delay_diff.py)
chm	(lidar_time, y, x)	float32	(1, 512, 512)	m; MCS_Lidar canopy height (build_mcs_datacube.py)
coherence	(pair, pol, y, x)	float32	(1, 1, 512, 512)	unitless [0,1]; ASF .cor.grd (build_mcs_datacube.py)
dsm	(lidar_time, y, x)	float32	(1, 512, 512)	m; MCS_Lidar surface DSM (build_mcs_datacube.py)
dswe	(pair, pol, y, x)	float32	(1, 1, 512, 512)	m w.e.; Guneriussen-2001 inversion, ρ=250, SNOTEL-anchored, pair 8 deramped (compute_dswe.py)
dtm	(lidar_time, y, x)	float32	(1, 512, 512)	m; MCS_Lidar bare-earth DTM (build_mcs_datacube.py)
lia	(line, y, x)	float64	(1, 351, 328)	degrees; JPL .llh/.lkv look vectors + 2023-02-09 DTM, corrected $\arccos(\hat{l} \cdot \hat{n})$ (2026-10-30) (compute_local_incidence_angle.py)
sd	(sd_time, y, x)	float32	(1, 512, 512)	m; QSI + MCS_Lidar snow depth, 0.5 m block-averaged to 3 m (build_mcs_datacube.py)
unwrapped_phase	(pair, pol, y, x)	float32	(1, 1, 512, 512)	radians; ASF .unw.grd; pairs 9/12/14/20/21 SNAPHU-unwrapped in-house (build_mcs_datacube.py, unwrap_missing_pairs.py)
wrapped_phase	(pair, pol, y, x)	complex64	(1, 1, 512, 512)	complex (unitless); ASF INTERFEROMETRY_GRD, complex-domain regrid (build_mcs_datacube.py)

Variable-level attributes record provenance: `unwrapped_phase` carries `custom_unwrapped_pairs = [9, 12, 14, 20, 21]` (pairs unwrapped in-house with SNAPHU because ASF ships no `.unw.grd`) and `dswe` carries the density assumption, mask rule, and SNOTEL anchoring description.

4 Dataset counts

Number of 2-D (y, x) maps per variable:

variable	2-D maps	composition
atm_delay	20	20 flight
atm_delay_diff	22	22 pair
chm	13	13 lidar_time
coherence	88	22 pair × 4 pol
dsm	13	13 lidar_time
dswe	88	22 pair × 4 pol
dtm	13	13 lidar_time
lia	2	2 line
sd	15	15 sd_time
unwrapped_phase	88	22 pair × 4 pol
wrapped_phase	88	22 pair × 4 pol

5 The `dswe` retrieval

ΔSWE (m water equivalent, positive = SWE gain from `pair_date1` to `pair_date2`) is retrieved per pair and polarization as follows: atmospheric correction $\varphi =$ `unwrapped_phase` − `atm_delay_diff`; for pair 8 a planar ramp (fit jointly with an elevation term that is deliberately *kept*) is removed; inversion via the Guneriussen et al. (2001) relation with the per-line local incidence angle and new-snow density $250\,\mathrm{kg\,m^{-3}}$; masking where coherence < 0.35 or $\cos(\mathrm{LIA}) \leq 0.05$; and anchoring of each map's unknown constant so the 90 m window at the Mores Creek Summit SNOTEL (637:ID:SNTL, 43.932°N, −115.666°E) matches the station's measured interval ΔSWE. Pairs 0, 1 and 13 could not be anchored (no coherent unwrapped data within 600 m of the station) and retain an arbitrary constant; the SNAPHU-unwrapped pairs may contain several connected components with independent constants. Validation (closure, cross-polarization, SNOTEL, lidar) is documented in `dswe_validation.md` and summarized in the companion report. Note: `lia` (and everything derived from it: `atm_delay`, `atm_delay_diff`, `dswe`) was regenerated on 2026-10-30 with the corrected $\arccos(\hat{l} \cdot \hat{n})$ formula, replacing the `uavsar_pytools calc_inc_angle` estimate, which is biased on slopes (SnowEx/uavsar_pytools issue #54); see `dswe_lia_fix_impact.md`.

6 Reading the store

Python (conda env `myenv`; see `environment.yml`):

```
import xarray as xr
ds = xr.open_zarr("ZARR/mores_creek_summit.zarr")
dswe = ds["dswe"].sel(pol="HH").isel(pair=8).load()   # one map, lazily chunked
```

Note: set `PROJ_DATA=/opt/anaconda3/envs/myenv/share/proj` before any reprojection work (see AGENTS.md). MATLAB has no Zarr reader on this system; use the NetCDF mirror `ZARR/mcs_gui.nc` (regenerated by `export_mcs_gui_nc.py`) or the interactive viewer `mcs_gui.m`.

7 The modeling-team subset: `mcs_top_pairs_subset.nc`

A 1.2 GB NetCDF4 file (regenerated by `export_subset_nc.py`) containing only what the snow modeling team needs, on the same 3 m grid:

- **pair** dimension of 6, in rank order — master-Zarr pairs 8, 2, 1, 5, 6 (ranks 1–5) and pair 4 (zero-change example); the `rank_note` coordinate labels each with the letters A–F used in the summary report's figure and tables. All six carry original ASF unwrapped phase (none needed custom unwrapping).
- Per pair: `dswe` (4 pols), `coherence` (4 pols, for quality screening), `unwrapped_phase` (4 pols) and `atm_delay_diff` (for transparency — already subtracted inside `dswe`).
- Site layers: `sd` for the two QSI lidar surveys (2020-02-09, 2021-03-15), `dem` (2023-02-09 lidar DTM), `lia` (both headings).
- Dates and labels are plain strings; all provenance is in the global attributes.

MATLAB: `ncread('mcs_top_pairs_subset.nc','dswe',[1 1 1 1],[Inf Inf 1 1])` reads the first pair's HH map (MATLAB returns dimensions reversed: x, y, pol, pair). Python: `xr.open_dataset(...)`. For interactive browsing, the delivered `mcs_subset_gui.m` (with `mcs_annotations.json` beside it and this file at `ZARR/mcs_top_pairs_subset.nc` or in the working directory) opens a four-panel linked viewer with pair/polarization/product dropdowns.